Assignment #2: Prolog Programming

Advait Vagerwal, Sahil Thakare, Elmoataz Abdellah

February 22, 2026

All members contributed to the assignment equally

1 State Representation

The state of the problem is represented as a list:

```
[ML, CL, Side]
```

where:

- `ML`: Number of missionaries on the **left** bank (0–3).
- `CL`: Number of cannibals on the **left** bank (0–3).
- `Side`: The location of the boat (`left` or `right`).

The initial state is defined as `[3, 3, left]` and the goal state is `[0, 0, right]`.

2 Implemented Search Algorithms

Two search procedures were implemented to solve the problem:

2.1 Depth-First Search (DFS) with Cycle Checking

The `solve_dfs(Path)` predicate explores the state space using a depth-first strategy. To prevent infinite loops in the search space, a `Visited` list is maintained. If a state has already been visited in the current path, the algorithm backtracks.

2.2 Breadth-First Search (BFS) for Shortest Path

The `solve_bfs(Path)` predicate implements a breadth-first search. This ensures that the first solution found is the one with the minimum number of river crossings. It uses a queue of paths to explore states level by level, also maintaining a visited list to avoid redundant computation and cycles.

3 How to Run

Follow these steps to execute the program and verify the results:

3.1 Loading and Running the Main Program

1. Start SWI-Prolog and load the program:

```
swipl -s mc.pl
```

2. To find a solution using **Depth-First Search**, execute:

```
?- run(dfs).
```

3. To find the **shortest solution** using **Breadth-First Search**, execute:

```
?- run(bfs).
```

The output will display each state in the path, the action (boat passengers) taken between states, and the total number of crossings.

3.2 Running the Test Suite

The project includes a comprehensive test file `tests.pl`. 1. Load the test file:

```
swipl -s tests.pl
```

2. Run all automated tests:

```
?- run_tests.
```

These tests verify safety constraints, move generation, path validity, and compare the path lengths between DFS and BFS.

4 Conclusion

The implementation successfully finds valid solution paths for the Missionaries and Cannibals problem. While both DFS and BFS reach the goal, BFS consistently provides the shortest path (11 crossings), whereas DFS may vary depending on the order of move generation.